

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_MCQ

Attempt : 2
Total Mark : 15
Marks Obtained : 14

Section 1 : MCQ

1. What will be the output of the following code?

```
public class Test {  
    public static void main(String[] args) {  
        int[] x = {4, 8, 12};  
        int result = x[0] * x[2];  
        System.out.println(result);  
    }  
}
```

Answer

48

Status : Correct

Marks : 1/1

2. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[][] arr = {  
            {5, 6, 7},  
            {8, 9, 10}  
        };  
        System.out.println(arr[0][2]);  
    }  
}
```

Answer

7

Status : Correct

Marks : 1/1

3. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3, 4};  
        for (int i = 0; i < a.length; i++) {  
            if (a[i] % 2 == 0)  
                a[i] = 0;  
        }  
        System.out.println(a[1] + " " + a[3]);  
    }  
}
```

Answer

0 0

Status : Correct

Marks : 1/1

4. What will be the output of the following code?

```
class Sample {  
    public static void main(String[] args) {
```

```

int[] a = {1, 2, 3};
int product = 1;
for (int i = 0; i < a.length; i++) {
    product *= a[i];
}
System.out.println(product);
}
}

```

Answer

6

Status : Correct

Marks : 1/1

5. What will be the output of the following code?

```

class Q {
    public static void main(String[] args) {
        int[][] a = {
            {1, 2},
            {3, 4}
        };
        for (int i = 0; i < a.length; i++) {
            for (int j = 0; j < a[0].length; j++) {
                System.out.print(a[i][j] + " ");
            }
        }
    }
}

```

Answer

1 2 3 4

Status : Correct

Marks : 1/1

6. What will be the output of the following code?

```

class M {
    public static void main(String[] args) {

```

```
int[][] arr = {
    {1, 2},
    {3, 4},
    {5, 6}
};

for (int i = 0; i < arr.length; i++) {
    System.out.print(arr[i][0] + " ");
}
}
```

Answer

1 3 5

Status : Correct

Marks : 1/1

7. What will be the output of the following code?

```
class Q {
    public static void main(String[] args) {
        int[][] a = {
            {1, 2},
            {3, 4}
        };
        int sum = 0;
        for (int i = 0; i < a.length; i++)
            for (int j = 0; j < a[0].length; j++)
                sum += a[i][j];
        System.out.println(sum);
    }
}
```

Answer

10

Status : Correct

Marks : 1/1

8. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] nums = {3, 6, 7, 2, 8};  
        int sum = 0;  
        for (int i = 0; i < nums.length; i++) {  
            if (nums[i] % 2 == 0)  
                sum += nums[i];  
        }  
        System.out.println(sum);  
    }  
}
```

Answer

16

Status : Correct

Marks : 1/1

9. What will be the output of the following code?

```
class Sample {  
    public static void main(String[] args) {  
        int[][] data = {  
            {1, 2},  
            {3, 4}  
        };  
        int sum = 0;  
  
        for (int[] row : data) {  
            for (int val : row) {  
                sum += val;  
            }  
        }  
  
        System.out.println("Sum = " + sum);  
    }  
}
```

Answer

Sum = 10

Status : Correct

Marks : 1/1

10. What will be the output of the following code?

```
class Sample {  
    public static void main(String[] args) {  
        int[][] matrix = {  
            {1, 2, 3},  
            {4, 5, 6}  
        };  
        System.out.println(matrix[1][2]);  
    }  
}
```

Answer

6

Status : Correct

Marks : 1/1

11. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3, 4};  
        for (int i = 0; i < a.length / 2; i++) {  
            int temp = a[i];  
            a[i] = a[a.length - 1 - i];  
            a[a.length - 1 - i] = temp;  
        }  
        System.out.println(a[0]);  
    }  
}
```

Answer

4

Status : Correct

Marks : 1/1

12. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] nums = {4, 2, 9, 5};  
        int max = nums[0];  
        for (int i = 1; i < nums.length; i++) {  
            if (nums[i] > max)  
                max = nums[i];  
        }  
        System.out.println(max);  
    }  
}
```

Answer

9

Status : Correct

Marks : 1/1

13. What will be the output of the following code?

```
class ReverseArray {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3, 4};  
        for (int i = 0; i < a.length / 2; i++) {  
            int temp = a[i];  
            a[i] = a[a.length - 1 - i];  
            a[a.length - 1 - i] = temp;  
        }  
        for (int i : a)  
            System.out.print(i + " ");  
    }  
}
```

Answer

4 3 2 1

Status : Correct

Marks : 1/1

14. What will be the output of the given code?

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3, 4, 5};  
        int n = arr.length;  
        int temp = arr[0];  
  
        for (int i = 0; i < n - 1; i++) {  
            arr[i] = arr[i + 1];  
        }  
        arr[n - 1] = temp;  
        for (int num : arr) {  
            System.out.print(num + " ");  
        }  
    }  
}
```

Answer

5 1 2 3 4

Status : Wrong

Marks : 0/1

15. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 1, 3, 1, 4};  
        int count = 0;  
        for (int i = 0; i < a.length; i++) {  
            if (a[i] == 1) count++;  
        }  
        System.out.println(count);  
    }  
}
```

Answer

3

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Rosh is intrigued by numerical patterns. Today, she stumbled upon a puzzle while working with arrays. She wants to compute the sum of the third-largest and second-smallest elements from a list of integers. She seeks your help to implement a program that solves this for her efficiently.

Input Format

The first line of input is an integer N, representing the size of the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output displays a single integer representing the sum of the third-largest and second-smallest elements in the array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10

10 20 30 40 50 60 70 80 90 100

Output: 100

Answer

```
// You are using Java
import java.util.Arrays;
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        Arrays.sort(arr);
        int third=arr[n-3];
        int sec=arr[1];
        System.out.println(third+sec);
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Monica is interested in finding a treasure but the key to opening is to get the sum of the main diagonal elements and secondary diagonal elements.

Write a program to help Monica find the diagonal sum of a square 2D array.

Note: The main diagonal of the array consists of the elements traversing from the top-left corner to the bottom-right corner. The secondary diagonal includes elements from the top-right corner to the bottom-left corner.

Input Format

The first line of input consists of an integer N, representing the number of rows and columns.

The following N lines consist of N space-separated integers, representing the 2D array elements.

Output Format

The first line of output prints "Sum of the main diagonal: " followed by an integer, representing the sum of the main diagonal.

The second line prints "Sum of the secondary diagonal: " followed by an integer, representing the sum of the secondary diagonal.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

4 5 6

7 8 9

Output: Sum of the main diagonal: 15

Sum of the secondary diagonal: 15

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        int[][] arr=new int[n][n];
```

```
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<n;j++){
```

```
                arr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int maindiagonal=0;
```

```
        int secdiagonal=0;
```

```
        for(int i=0;i<n;i++){
```

```
            maindiagonal+=arr[i][i];
```

```
            secdiagonal+=arr[i][n-i-1];
```

```
        }
```

```
System.out.println("Sum of the main diagonal: "+maindiagonal);  
System.out.println("Sum of the secondary diagonal: "+secdiagonal);  
}  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

You are developing a warehouse management system for a shipping company. The system uses an integer array to represent the weights of packages in a specific order. To verify that the weight capacity is not exceeded, the program needs to calculate the sum of the weights of the first and last packages in the list.

Task:

Write a code to calculate the sum of the weights of the first and last packages in the list. The program should take an integer array as input and return the total weight of the first and last packages.

Input Format

The first line of the input is an integer N representing the size of the array.

The second line of the input is N space-separated integer values.

Output Format

The output is displayed in the following format:

"Sum of the first and last elements: <<Sum>>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: Sum of the first and last elements: 60

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        int[] arr=new int[n];
```

```
        for(int i=0;i<n;i++){
```

```
            arr[i]=sc.nextInt();
```

```
        }
```

```
        int sum=arr[0]+arr[n-1];
```

```
        System.out.println("Sum of the first and last elements: "+sum);
```

```
    }  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int m=sc.nextInt();
        int[][] arr=new int[n][m];
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                arr[i][j]=sc.nextInt();
            }
        }
        int[][] brr=new int[n][m];
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                brr[i][j]=sc.nextInt();
            }
        }
    }
}
```

```
    }  
    }  
    int[][] res=new int[n][m];  
    for(int i=0;i<n;i++){  
        for(int j=0;j<m;j++){  
            res[i][j]=arr[i][j]+brr[i][j];  
        }  
    }  
    for(int i=0;i<n;i++){  
        for(int j=0;j<m;j++){  
            System.out.print(res[i][j]+" ");  
        }  
        System.out.println();  
    }  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sharon is creating a program that finds the first repeated element in an integer array. The program should efficiently identify the first element that appears more than once in the given array. If no such element is found, it should appropriately display a message.

Help Sharon to complete the program.

Input Format

The first line of input consists of an integer n, representing the number of elements in the array.

The second line consists of n space-separated integers, representing the array elements.

Output Format

If a repeated element is found, print the first element that appears more than once.

If no repeated element is found, print "No repeated element found in the array".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 8

12 21 13 14 21 36 47 21

Output: 21

Answer

// You are using Java

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }

        boolean found=false;
        for(int i=0;i<n-1;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]==arr[j]){
                    System.out.print(arr[i]);
                    found=true;
                    break;
                }
            }
        }
        if(found){
            break;
        }
    }
}
```

```
}  
    if(!found){  
        System.out.println("No repeated element found in the array");  
    }  
}  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_PAH

Attempt : 1
Total Mark : 40
Marks Obtained : 37.5

Section 1 : Coding

1. Problem Statement

Eminem is a billiard player who enjoys playing billiards and also likes solving mathematical puzzles. He notices that the billiard balls on the table are arranged in a grid, and he is curious to find the sum of the numbers written on each ball.

Write a program to find the sum of all the numbers written on each ball in the grid.

Input Format

The first line of input consists of an integer N, representing the number of rows.

The second line consists of an integer M, representing the number of columns.

The following lines N lines consist of M space-separated integers, representing the numbers written on each ball.

Output Format

The output prints an integer representing the sum of all the numbers written on each ball.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

3

1 2 3

4 5 6

7 8 9

Output: 45

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        int m=sc.nextInt();
```

```
        int[][] arr=new int[n][m];
```

```
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<m;j++){
```

```
                arr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int sum=0;
```

```
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<m;j++){
```

```
                sum+=arr[i][j];
```

```
            }
```

```
        }
```



```
        System.out.print(sum);  
    }  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Egath is participating in a coding hackathon, and one of the challenges requires him to work with an array of integers. The task is to remove exactly one element from the array such that the sum of the remaining elements is a prime number.

Help Egath find the first possible prime sum by removing one element or determining if no such prime sum can be achieved.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line consists of N space-separated integers, representing the array elements.

Output Format

If removing one element results in a prime sum, print the sum.

If no such prime sum can be achieved by removing exactly one element, print "No valid prime sum found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: 5

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        int t=0;
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
            t+=arr[i];
        }
        boolean found=false;
        for(int i=0;i<n;i++){
            int sum=t-arr[i];
            if(isPrime(sum)){
                System.out.println(sum);
                found=true;
                break;
            }
        }
        if(!found){
            System.out.println("No valid prime sum found");
        }
    }
    public static boolean isPrime(int n){
        if(n<=1) return false;
        if(n==2) return true;
        if(n%2==0) return false;
        for(int i=3;i<=Math.sqrt(n);i+=2){
            if(n%i==0) return false;
        }
        return true;
    }
}
```

Status : Correct**Marks : 10/10**

3. Problem Statement

Priya is building a system to automate image transformations using matrix operations. To do this, she needs to multiply two matrices representing pixel data and transformation rules.

Help Priya perform matrix multiplication and print the resulting matrix if the operation is valid.

Input Format

The first line of input consists of two int values, representing the number of rows $R1$ and columns $C1$ of the first matrix.

The next $R1 \times C1$ integers represent the elements of the first matrix.

The next line consists of two int values, representing the number of rows $R2$ and columns $C2$ of the second matrix.

The next $R2 \times C2$ integers represent the elements of the second matrix.

Output Format

If matrix multiplication is possible, print $R1$ lines, each containing $C2$ space-separated int values representing the resulting matrix.

Otherwise, print "Matrix multiplication not possible".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2 3

1 2 3

4 5 6

3 2

7 8

9 10

11 12

Output: 58 64

139 154

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n1=sc.nextInt();
```

```
        int m1=sc.nextInt();
```

```
        int[][] arr=new int[n1][m1];
```

```
        for(int i=0;i<n1;i++){
```

```
            for(int j=0;j<m1;j++){
```

```
                arr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int n2=sc.nextInt();
```

```
        int m2=sc.nextInt();
```

```
        int[][] brr=new int[n2][m2];
```

```
        for(int i=0;i<n2;i++){
```

```
            for(int j=0;j<m2;j++){
```

```
                brr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int[][] res=new int[n1][m2];
```

```
        for(int i=0;i<n1;i++){
```

```
            for(int j=0;j<m2;j++){
```

```
                for(int k=0;k<m1;k++){
```

```
                    res[i][j]+=arr[i][k]*brr[k][j];
```

```
                }
```

```
            }
```

```
        }
```

```
        for(int i=0;i<n1;i++){
```

```
            for(int j=0;j<m2;j++){
```

```
                System.out.print(res[i][j]+(m2==j-1 ? "" : " "));
```

```
            }
```

```
        System.out.println();
```

```
    }
```

```
}
```

}

Status : Partially correct

Marks : 7.5/10

4. Problem Statement

In a customer loyalty program, reward points are logged in a sorted array as customers make transactions. Occasionally, due to system errors, duplicate entries for the same transaction may appear. To ensure accurate reward calculations, it's crucial to remove these duplicates from the list.

Write a program to process the array of reward points, removing any duplicates while preserving the order of unique entries. The program should then display the cleaned list of unique reward points and the total count of these unique points.

Input Format

The first line of input consists of an integer N, representing the number of reward points.

The second line consists of N space-separated integers, representing the reward points in sorted order.

Output Format

The first line of output prints the cleaned list of unique reward points separated by a space.

The second line of output prints an integer representing the total count of unique reward points.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

100 100 200

Output: 100 200

2

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        int c=0;
        for(int i=0;i<n;i++){
            if(i==0||arr[i]!=arr[i-1]){
                System.out.print(arr[i]+" ");
                c++;
            }
        }
        System.out.println();
        System.out.println(c);
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Robin is a tech-savvy teenager who is diving into programming.

He is working on a project to find special elements in an array called 'leaders.' Leaders are those exceptional elements that are greater than the sum of all the elements to their right.

Assist Robin in writing this program.

Example

Input:

6

16 28 74 19 25 11

Output:

74 25 11

Explanation:

The element 16 is not greater than the sum of elements to its right ($28 + 74 + 19 + 25 + 11 = 157$)

The element 28 is not greater than the sum of elements to its right ($74 + 19 + 25 + 11 = 129$)

The element 74 is greater than the sum of elements to its right ($19 + 25 + 11 = 55$)

The element 19 is not greater than the sum of elements to its right ($25 + 11 = 36$)

The element 25 is greater than the sum of elements to its right (11)

The last element 11 is always a leader since there are no elements to its right.

So, the output is {74, 25, 11}.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line consists of N space-separated integers, representing the elements of the array.

Output Format

The output prints the special elements in the given array, that are greater than the sum of all the elements to their right.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

3 4 2 5 1

Output: 5 1

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        for(int i=0;i<n;i++){
            int r=0;
            for(int j=i+1;j<n;j++){
                r+=arr[j];
            }
            if(arr[i]>r){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Nikila is working as an intern in a software firm and is practicing with a matrix where each row represents a set of numerical values. Her task is to identify the row with the highest sum of its elements and remove that row from the matrix. After removing the row with the highest sum, Nikila needs to print the updated matrix.

Your task is to help Nikila in implementing the same. If there are two or more rows that have same the highest sum, the firstly encountered row is deleted.

Input Format

The first line of the input consists of two space-separated integers, R and C, representing the number of rows and columns in the matrix, respectively.

The following R lines each contain, C space-separated integers representing the matrix elements.

Output Format

The output prints the matrix after removing the row with the highest sum. Each row should be printed on a new line, with elements separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 2 2

1 2

3 4

Output: 1 2

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int r=sc.nextInt();
```

```
        int c=sc.nextInt();
```

```
        int[][] arr=new int[r][c];
```

```
        for(int i=0;i<r;i++){
```

```
            for(int j=0;j<c;j++){
```

```
                arr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int max=Integer.MIN_VALUE;
```

```
        int row=-1;
```

```
        for(int i=0;i<r;i++){
```

```
            int sum=0;
```

```
            for(int j=0;j<c;j++){
```

```
                sum+=arr[i][j];
```

```
            }
```

```

        if(sum>max){
            max=sum;
            row=i;
        }
    }
    for(int i=0;i<r;i++){
        if(i==row) continue;
        for(int j=0;j<c;j++){
            System.out.print(arr[i][j]+" ");
        }
        System.out.println();
    }
}
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement:

Emma, a budding computer vision enthusiast, is working on a challenging image processing project. She has a square image represented as a 2D matrix of integers. As part of a special filter operation, she needs to rotate the image by 90 degrees clockwise, but there's a twist — she must perform the rotation in-place, using no extra space.

This means Emma has to rotate the matrix without creating a new one. Your task is to help her implement a Java program that takes this square matrix as input and rotates it within the same structure.

Can you help Emma efficiently rotate the image so that her project can move to the next stage?

Input Format

The first line of input contains a single integer n , representing the number of rows and columns of the square matrix (i.e., the matrix is of size $n \times n$).

The next n lines each contain n space-separated integers, representing the elements of each row of the 2D array.

Output Format

The first line of output prints "Rotated 2D Array:"

The next n lines of output print the rotated matrix.

Each line contains n space-separated integers representing a row of the rotated matrix.

Refer to the sample output for format specification.

Sample Test Case

Input: 3

1 2 3

4 5 6

7 8 9

Output: Rotated 2D Array:

7 4 1

8 5 2

9 6 3

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        int[][] arr=new int[n][n];
```

```
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<n;j++){
```

```
                arr[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int[][] res=new int[n][n];
```

```
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<n;j++){
```

```
                res[j][n-1-i]=arr[i][j];
```

```
            }
```

```
        }
```

```

System.out.println("Rotated 2D Array:");
for(int i=0;i<n;i++){
    for(int j=0;j<n;j++){
        System.out.print(res[i][j]+" ");
    }
    System.out.println();
}
}
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Rina is managing the inventory for a library, where each row of a 2D matrix represents the number of different genres of books available on each shelf.

She wants to perform the following operations:

Transformation: Replace each element in a row with the sum of all elements in that row.
Merging: After transformation, Rina will provide one additional matrix, and specify whether to merge the transformed matrix with this new matrix row-wise or column-wise.

Input Format

The first line contains two integers R and C, representing the number of rows and columns of the initial matrix.

The next R lines contain C space-separated integers, representing the book counts in the library.

The next line contains two integers MR and MC, representing the dimensions of the second matrix (to be merged).

The next MR lines contain MC space-separated integers, representing the second matrix.

The last line contains an integer mergeType:

- 0 Row-wise merging (append the second matrix below the transformed matrix).

- 1 Column-wise merging (append the second matrix to the right of the transformed matrix).

Output Format

The output prints "Transformed matrix: " followed by the transformed 2D matrix where each element in a row is replaced with the sum of the elements in that row.

The output prints "Final merged matrix: ", followed by the merging based on mergeType.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 4

8 2 4 9

4 5 6 1

7 8 9 3

2 4

3 5 7 2

6 1 4 9

0

Output: Transformed matrix:

23 23 23 23

16 16 16 16

27 27 27 27

Final merged matrix:

23 23 23 23

16 16 16 16

27 27 27 27

3 5 7 2

6 1 4 9

Answer

```
// You are using Java
import java.util.Scanner;
import java.util.Arrays;
class Main{
    public static void main(String[] args){
```

```

Scanner sc=new Scanner(System.in);
int r=sc.nextInt();
int c=sc.nextInt();
int[][] arr=new int[r][c];
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        arr[i][j]=sc.nextInt();
    }
}
for(int i=0;i<r;i++){
    int rowsum=0;
    for(int j=0;j<c;j++){
        rowsum+=arr[i][j];
    }
    Arrays.fill(arr[i],rowsum);
}
int mr=sc.nextInt();
int mc=sc.nextInt();
int[][] sec_arr=new int[mr][mc];
for(int i=0;i<mr;i++){
    for(int j=0;j<mc;j++){
        sec_arr[i][j]=sc.nextInt();
    }
}
int mergeType=sc.nextInt();
System.out.println("Transformed matrix:");
print(arr);
System.out.println("Final merged matrix:");
if(mergeType==0){
    int[][] merged=new int[r+mr][];
    for(int i=0;i<r;i++)
        merged[i]=arr[i];
    for(int i=0;i<mr;i++)
        merged[r+i]=sec_arr[i];
    print(merged);
}
else{
    int[][] merged=new int[r][c+mc];
    for(int i=0;i<r;i++){
        for(int j=0;j<c;j++)
            merged[i][j]=arr[i][j];
    }
}

```

```
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                merged[i][c+j]=sec_arr[i][j];
            }
            print(merged);
        }
    }
    public static void print(int[][] mat){
        for(int[] row:mat){
            for(int val:row){
                System.out.print(val+" ");
            }
            System.out.println();
        }
    }
}
```

Status : Correct

Marks : 10/10